

Research Exploration

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Some Good Resources for NNDL

- Elizabeth DePoy, "Introduction to Research", 7th Edition, Elsevier, 2015.
- Arpita Baijal, "Introduction to Research", Lambert Academic Publishing.
- C. R. Kothari, "Research methodology", New Age International Publishers, 2004.
- <https://advice.writing.utoronto.ca/types-of-writing/literature-review/>
- What is a Literature Review? - Research Guides - WashU
- B. Yegnanarayana, "Evolving Ideas of Teaching and Research in Engineering Institutions During My Journey in Academics Over Five Decades" The Mind of an Engineer: Volume 2, 2020



Organization

- What is a research exploration?
- Literature review & ready
- Taking up first & exploring
- What is involved in research exploration?
- How to go about?
- Significance of hypothesis
- Experiments or theory proof
- Expected line vs opposite
- Results and discussion
- Summary and conclusions

What is a literature review?

- A literature review discusses and analyses published information in a particular subject area.
- A **summary** is a recap of the important information of the source.
- A **synthesis** is a re-organization, or a reshuffling, of that information.
- New interpretation of old material.
- It might trace intellectual progression of field.
- Literature review may evaluate sources and advise reader on the most pertinent or relevant.

Significance of Literature Review

- Literature review provides a handy guide to a particular topic.
- If you have limited time to conduct research, literature review can give you an overview or act as a stepping stone.
- Literature review also provides a solid background for a research paper's investigation.
- Comprehensive knowledge of the literature of the field is essential to identify gaps and hypothesize new directions.
- What knowledge and ideas have been established on a topic (**review**), and what their strengths and weaknesses are, accordingly, scope for future work (**critical review**).

Getting Ready for Research Exploration: Plan A

- Identified broad area of research in consultation with supervisor
- Done set of good courses in consultation with supervisor and successfully completed
- Completed comprehensive examination, if any
- Identified a narrow research topic or direction from the initial broad area in consultation with supervisor
- Done literature review and prepared first draft of document
- Identified gaps in existing literature review based on Critical Review
- Ready to launch Research Exploration



Getting Ready for Research Exploration: Plan B

- Done set of courses in consultation with supervisor
- Completed comprehensive examination, if any
- Supervisor suggested a narrow research topic
- Supervisor suggests a ready problem to explore
- Done brief literature review to support hypothesis
- Ready to launch Research Exploration
- Beware of overfitting to supervisor ideas and suggestions
- Problem of local minima and fear to losing generalization
- Threat of not becoming an independent researcher



Scholar & Supervisor Relationship

- Spirit of collaboration and joint exploration
- An experienced researcher and a novice
- **Plan A:** Looks like slow path, but good path to become independent researcher
- **Plan B:** Looks like fast path, but need more time to become independent researcher
- Both are ok, but decide based on what you are intending to achieve as part of research degree
- It also depends on supervisor
- 80% is scholar's effort and 20% is supervisor effort
- Be conscious of becoming independent researcher capabilities



Ready to Launch Research Journey: Thumb Rule

- Typically 5-6 research gaps for PhD $\implies$ at least 3 novel results $\implies$ 2-3 journal articles
- Typically 3-4 research gaps for MTech(R) $\implies$ at least 2 novel results $\implies$ 1-2 journal articles
- Typically 2-3 research gaps for UGR $\implies$ at least one novel result $\implies$ 1-2 conference / journal articles
- May be you can start research exploration as soon as you identify one research gap, but keep hunting for more gaps as suggested above

Launching First Research Exploration

- Clarity on Research Hypothesis
- Existing gap is this ...
- Hypothesize that can be explored - novel direction may emerge
- Clarity in mind is important (scholar or supervisor or both)
- Setting up theoritical framework and exploring new proofs
- Setting up experiment facility and carrying out experiments
- Setting up simulation framework and implementing hypothesis
- Writing Research Diary is useful (100 - 150 words)



Research Exploration: Outcome Possibilities

- **All is well:** Results according to hypothesis
- **Not so well:** Results half-way and need refinement of hypothesis
- **Wrong Hypothesis :** No results, fallacy in hypothesis and drop
- **Opposite Hypothesis:** Exactly opposite to the hypothesis

Research Exploration: All is well

- Sit in desk, plan experiments
- Perform experiments as planned, collect results
- Tabulate results
- Generate figures
- Comeback to desk, write research diary
- Think about any missing links to justify your hypothesis
- Repeat above steps until you see a completeness in planned effort
- Typically 3-6 months effort

Research Exploration: Not so well

- Sit in desk, plan experiments
- Perform experiments as planned, collect results
- Tabulate results & generate figures
- Comeback to desk, write research diary
- Looks like results are not along expected directions
- Did not think enough to make it general / from particular limitation
- Think to modify of your hypothesis
- Repeat above steps until you see a completeness in planned effort
- Typically 3-6 months effort



Research Exploration: Wrong Hypothesis

- Sit in desk, plan experiments
- Perform experiments as planned, collect results
- Tabulate results & generate figures
- Comeback to desk, write research diary
- Looks like results are not at all coming
- Think why it is so?
- Analyze why it is so.
- Repeat above steps until you are convinced it is a wrong hypothesis
- Typically 3-6 months effort, move on to new peace of work



Research Exploration: Opposite Hypothesis

- Sit in desk, plan experiments
- Perform experiments as planned, collect results
- Tabulate results & generate figures
- Comeback to desk, write research diary
- Looks like results are exactly opposite what you hypothesized
- Think why it is so?
- Analyze why it is so.
- Repeat above steps until you are convinced the results are correct and then agree to the opposite hypothesis
- Typically 3-6 months effort, move on to new peace of work



Summary of Research Exploration

- From literature identified few leads
- Ready to launch research exploration
- Do the research exploration
- One of the possibilities arrived
- Enough novel contributions
- Ready for writing research article or patent

Thank You